AI POWERED INTELLIGENT HOME PLANNING & OPTIMIZATION SYSTEM

AI-Based EXTERIOR SPACE OPTIMIZATION

Project ID: 25-26J-201

BSc (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

April 2026

AI POWERED INTELLIGENT HOME PLANNING & OPTIMIZATION SYSTEM

AI-Based EXTERIOR SPACE OPTIMIZATION

Project ID: 25-26J-201

BSc (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

April 2026

Declaration

I declare that this is my own work and this dissertation does not incorporate without acknowledgment any material previously submitted for a Degree or Diploma in any other University or institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due reference is made in the text.

I hereby grant to the Sri Lanka Institute of Information Technology (SLIIT) the non-exclusive right to reproduce and distribute my dissertation, in whole or in part, in print, electronic or other media. I retain the right to use this content in whole or in part in future works (such as articles or books).

Signature:

Date : 23/04/2026

Abstract

Urban housing in Sri Lanka is increasingly facing the challenge of shrinking external spaces such as gardens, rooftops, and facades. Traditional design approaches often emphasize aesthetics and ignore environmental conditions, cultural traditions, and user-specific needs. As a result, the exterior layouts of urban homes are often underutilized, energy inefficient, and less sustainable. This project proposes an AI-powered system for outdoor space optimization and visualization that integrates environmental data and user preferences to generate functional and sustainable outdoor designs. The system will use data such as solar exposure, wind direction, soil conditions and local climate patterns to optimize tree placement, roof use and facade design. Generative AI models will be used to suggest layouts that are culturally relevant and ecologically adaptive. Photorealistic visualization tools based on ControlNet and diffusion models allow users to instantly preview design changes. This methodology involves collecting spatial and environmental data, applying geospatial and generative AI techniques to produce optimal layouts, and simulating energy impacts through solar shading and airflow models. A user-friendly interface enables homeowners to customize their designs in real time, while monitoring aesthetic and environmental results. The desired result is a dynamic tool that empowers Sri Lankan homeowners to make informed design decisions that balance modern sustainability with traditional architectural aesthetics. By promoting passive cooling, microclimate regulation, and effective space use, this project supports sustainable urban living and enhances outdoor comfort in environments with limited space.

Keywords: Exterior optimization, AI design, sustainability, visualization, Sri Lanka

Table of Contents

List Of Figures

1. INTRODUCTION

Urban housing in Sri Lanka is increasingly facing challenges due to the rapid shrinkage of outdoor spaces such as gardens, rooftops, and facades due to population growth, land scarcity, and urban density. These outdoor spaces are more than just aesthetic additions — They contribute to the preservation of cultural practices such as natural ventilation, passive cooling, microclimate regulation, recreation, food cultivation, and layouts inspired by material science. When neglected or poorly designed, outdoor spaces not only lose their functional and cultural value, but also contribute to increased indoor energy consumption, reduced comfort, and unsustainable urban living conditions.

Traditional approaches to exterior design are aesthetically driven and fail to integrate critical environmental and cultural considerations. Existing tools rarely incorporate solar paths, wind direction, soil quality, or microclimate data into the design process. They also lack interactivity and real-time visualization, limiting homeowners' ability to explore and evaluate design options. As a result, the exterior layouts of urban housing in Sri Lanka are often underutilized, energy inefficient, and disconnected from both environmental realities and cultural expectations.

Recent advances in artificial intelligence, computer vision, and generative design have created new opportunities to address these limitations. Geospatial AI techniques can analyze terrain and environmental features, while generative AI models like ControlNet and diffusion-based renderers enable photo-realistic, real-time previews of proposed designs. These technologies, combined with simulation models for solar shading, wind flow, and energy impacts, provide a way to create exterior spaces that are not only visually appealing but also energy efficient, sustainable, and culturally relevant.

Therefore, this project proposes an AI-powered outdoor space optimization and visualization system that dynamically integrates environmental data, user preferences, and cultural values to maximize the usability of shrinking outdoor spaces. The system will optimize garden layouts, tree placements, facade, and roof configurations, and provide interactive and photo-realistic design previews.

The expected main contributions are fourfold:

1. A novel AI-based optimization tool for exterior spaces tailored to the urban context of Sri Lankan.

2. Real-time photo-realistic visualizations to support interactive design decisionmaking.

3. Enhanced energy efficiency and passive cooling through optimal greening and facade placement.

4. A sustainable design framework that balances modern urban needs with traditional architectural aesthetics.

By aligning sustainability, technology, and culture, this project aims to provide Sri Lankan homeowners with an intelligent decision-making support tool. It transforms outdoor spaces into functional, ecologically adaptive, and culturally meaningful extensions.

1.1. Background & Literature Survey

Due to rapid urbanization, rising land prices, and high population density, Sri Lanka's urban housing sector is facing a growing crisis of shrinking outdoor spaces such as gardens, rooftops, and facades. Traditionally, these spaces are much more than decorative elements. In the Sri Lankan context, gardens and outdoor areas have long been associated with cultural values, food production, passive cooling, natural ventilation, and community interaction. Families often relied on small gardens not only for aesthetic purposes, but also to grow fruits, vegetables, and medicinal plants that directly contributed to food security and community well-being. The principles of materialism, the traditional design philosophy of Sri Lanka, emphasize harmony between built structures, natural elements, and environmental forces [6]. According to these traditions, house orientation, tree placement, and integration of water and open spaces were carefully planned to maintain energy balance, airflow, and spirituality within a home.

However, as modern urban housing expands, these cultural and environmental functions are increasingly being ignored or minimized. Outdoor spaces are often reduced to narrow balconies, decorative facades, or small green patches, which contribute little to energy efficiency, sustainability, or indoor well-being. This has led to a situation where outer areas are underutilized and disconnected from both environmental realities and traditional cultural practices [7]. Without proper planning, these reduced outdoor areas can contribute to increased indoor temperatures, reduced ventilation, increased reliance on artificial cooling, and an overall decline in urban living standards.

Globally, considerable progress has been made in the application of artificial intelligence (AI) to landscape and urban design. Surveys such as Xing et al. [1] Provides comprehensive reviews of how machine learning, deep learning, computer vision, and generative AI are transforming landscape architecture. These technologies have been applied to tasks such as urban forest assessment (i-Tree Eco), automatic plant recognition using CNN and FCN, irrigation optimization systems, and AI-assisted master planning. AI enables designers and planners to make data-driven decisions, considering vast amounts of environmental, social, and spatial data that are difficult to process manually.

Several recent works demonstrate the potential of generative AI in transforming landscape visualization and planning. For example, planetography allows for iterative landscape visualization using large language models (LLMs) and diffusion-based rendering, moving from conceptual cues to photorealistic creations in a phased manner [2]. Ubangian, on the other hand, combines segmentation networks and diffusion models to reconstruct urban green spaces at the city scale, enabling urban planners to visualize and modify large-scale green infrastructure [3]. Similarly, Layout2Rendering combines deep learning with grassroots parametric modeling tools to transform design layouts into detailed 3D renderings, this allows designers to quickly produce interactive prototypes that can be adapted to different contexts [4]. Ashari et al. compared AI-based rendering tools such as stable diffusion, control networks, and LoRA with traditional workflows, highlighting how AI achieves superior visual fidelity and efficiency [5].

These developments show that AI has enormous potential in the areas of design automation, visualization, and optimization, but most of these systems are focused on large-scale planning, urban regeneration, or general-purpose landscape design. Their aim is to assist city planners, architects, and policymakers in managing large urban areas, parks, or ecological corridors. Few of these studies focus specifically on the optimization of outdoor space at the household level, where the constraints of space, climate, and cultural values are most critical.

Existing systems emphasize visual realism, automated design generation, or large-scale environmental monitoring, and often measure success using visual metrics such as SSIM or FID scores rather than sustainable outcomes. They rarely address how the small, limited exterior spaces of dense urban housing can be intelligently optimized to maximize functionality, sustainability, and cultural identity. Furthermore, these systems typically do not integrate local climate data, microclimate simulations, or cultural traditions into their design processes.

This leaves a clear research gap for solutions tailored to the Sri Lankan context. Residential outdoor spaces, while shrinking, are essential for comfort, energy efficiency, cultural practices, and sustainable living. An AI-powered framework that combines environmental data (sunlight exposure, wind direction, soil quality), user preferences (plant choices, recreational needs), and cultural traditions (Wasthu Vidya principles) is desperately needed to create exterior designs that are sustainable, optimized, and culturally significant. In the face of rapid urban growth, such a framework would enhance energy efficiency, climate resilience, and urban sustainability, in addition to preserving the tradition and identity of Sri Lankan living spaces.

1.2. Research Gap

Even though research from around the world has demonstrated how artificial intelligence (AI) can revolutionize landscape architecture and urban planning, there are still a lot of unanswered questions when it comes to applying these methods in Sri Lanka, especially at the household level where outdoor spaces are limited, culturally significant, and small.

A major gap is the integration of environmental data into AI-based planning systems. Many existing AI visualization and design tools, such as Planetography, Ubangian, and Layout2Rendering, focus heavily on visual aesthetics, rendering fidelity, or large-scale reproducibility [2–4]. While these tools can generate photorealistic designs, they often do so without considering critical environmental factors such as sun exposure, wind direction, precipitation patterns, soil conditions, or microclimate regulation. In Sri Lanka's tropical climate, where solar intensity and ventilation strongly affect indoor comfort and energy demand, this lack of environmental considerations results in designs that may look attractive but are not sustainable or functional.

The existing AI-driven design systems' lack of cultural adaptation is another significant gap. Wasthu Vidya, which establishes guidelines for spatial harmony, orientation, and the symbolic placement of both natural and constructed features, has influenced traditional Sri Lankan housing architecture [6]. But none of the AI systems examined in international literature specifically take traditional or cultural frameworks into account. This causes a gap between Sri Lankan households' cultural identities and technologically sophisticated designs, which makes such solutions less feasible or acceptable for local populations.

The current technologies' minimal user control and interactivity represent a third gap. Although automated outputs are produced by models such as Layout2Rendering and Ubangian, they provide limited opportunities for real-time user customization [3][4]. In Sri Lanka, homeowners and designers hardly ever get the chance to test out various garden or rooftop configurations, interactively modify layouts, or instantly see how these decisions impact environmental performance (e.g., shading, ventilation, or space use). The lack of interactive tools makes the design process more technologist-driven than userdriven, which lessens its applicability to homes that require highly customized and flexible solutions.

Another significant problem is the scale discrepancy between current research and Sri Lankan needs. Numerous AI systems are created for institutional projects, city redevelopment, or urban-scale planning [1][3]. UrbanGenAI, for instance, is quite good at reconstructing landscapes at the city level, but it is not made to manage the complexities of a small residential facade, a rooftop patio, or a 10x10 m urban garden. Because each square meter in Sri Lanka's urban housing environment must fulfill several functions, including ventilation, recreation, gardening, and cultural expression, these tiny areas demand meticulous optimization. The existing corpus of AI research mainly ignores the necessity of micro-scale optimization.

Finally, most AI-driven landscape systems have a performance evaluation gap. Research findings frequently emphasize user impression of aesthetics or visual accuracy measures (e.g., SSIM, FID, LPIPS) [5]. Although useful for visualization, these do not quantify sustainable results like biodiversity contributions, passive airflow enhancements, or decreases in indoor cooling loads. The lack of sustainability-focused evaluation measures significantly restricts the practicality of current methods in Sri Lanka, where energy efficiency and climate resilience are urgent issues.

The primary research gaps, in brief, are:

- Environmental integration gap: AI-driven exterior design lacks data on soil, wind, solar, and microclimate [2–4].
- Cultural adaptability gap: generative design does not incorporate Wasthu Vidya or traditional values [6].
- The inability of homeowners to alter and observe designs in real time is known as the "user interactivity gap" [3][4].
- Scale gap: current AI systems aren't designed for small home exteriors, but rather for large-scale planning [1][3].
- Sustainability evaluation gap: existing systems prioritize aesthetics above ecological performance or energy efficiency [5].
- Lack of AI-powered solutions created especially for Sri Lankan urban housing situations is known as the "local context gap" [7].

1.3. Research Problem

Due to urbanization, land scarcity, and high population density, outside areas including gardens, rooftops, and facades are fast decreasing in Sri Lankan urban housing [7]. These outdoor spaces, which were originally essential to Sri Lankan culture, have long supported Wasthu Vidya-guided cultural customs, natural ventilation, food cultivation, leisure, and cooling [6]. Modern home complexes, however, are placing a greater emphasis on building economy and aesthetics than on cultural significance and environmental sustainability. Because of this, these outdoor designs are frequently underutilized, energyinefficient, and disengaged from both ecological realities and conventional behaviors.

Advanced AI-based design solutions have been introduced by international research, but they are still insufficient for the particular requirements of Sri Lankan dwellings. While existing technologies like PlantoGraphy, UrbanGenAI, and Layout2Rendering show promise in generative visualization and design automation [2–4], their primary focus is on institutional landscapes or large-scale urban planning [1][3]. Household-level limitations, where outdoor spaces are extremely limited and demand precise optimization to balance numerous tasks, are rarely addressed by these systems.

A number of fundamental issues surface:

- Inadequate environmental integration: In Sri Lanka's tropical climate, sun pathways, wind dynamics, soil quality, and microclimate data are all crucial, yet current AI systems do not include them in design outputs [2–4].

- Cultural disconnect: Because current AI tools disregard Wasthu Vidya principles, they produce designs that might be aesthetically pleasing but do not reflect regional customs and values [6].

- Limited user interaction: Because homeowners find it difficult to alter or evaluate designs in real time, solutions are primarily driven by technologists rather than by users [3][4].

- Why Only pay attention to visual metrics: Existing assessments place a strong emphasis on FID, SSIM, and rendering quality, but they do not account for sustainability effects like better airflow, lower cooling demand, or biodiversity advantages [5].

- Scale mismatch: The majority of systems are made for urban-scale projects and aren't tailored to maximize the tiny outside areas seen in Sri Lankan homes [1][3].

Research Problem Statement:
How can an AI-powered system that integrates environmental data (sunlight, wind, soil, climate), cultural values (Wasthu Vidya), and user preferences be created to optimize the shrinking exterior spaces in urban homes in Sri Lanka? The system should provide realtime photorealistic visualization to support design decisions that are sustainable, energyefficient, and culturally significant.

2. OBJECTIVES

2.1. Main Objective

In order to support sustainable, energy-efficient, and culturally significant design choices, an AI-powered system that integrates environmental data, cultural values, and user preferences will optimize the shrinking exterior spaces of Sri Lankan urban homes, including gardens, rooftops, and facades. The system will also provide real-time photorealistic visualizations.

2.2. Specific Objectives

- To improve energy efficiency and microclimate regulation, incorporate environmental data (solar exposure, wind direction, soil conditions, rainfall, and local climate) into the design optimization process.
- To make sure external plans are in line with Sri Lankan architectural ideals and user acceptance, use cultural customs like Wasthu Vidya.
- To suggest clever arrangements for trees, garden designs, rooftops, and facades in limited urban areas, create AI-driven optimization models (geospatial AI + generative AI).
- Give consumers the ability to see and modify their external designs instantaneously using real-time interactive visualizations that employ ControlNet and diffusion-based rendering.
- To evaluate design performance beyond aesthetics, set up sustainability-focused evaluation indicators (such as energy savings, biodiversity contribution, ventilation enhancement, and passive cooling effectiveness).

3. METHODOLOGY

3.1. System Overview Diagram

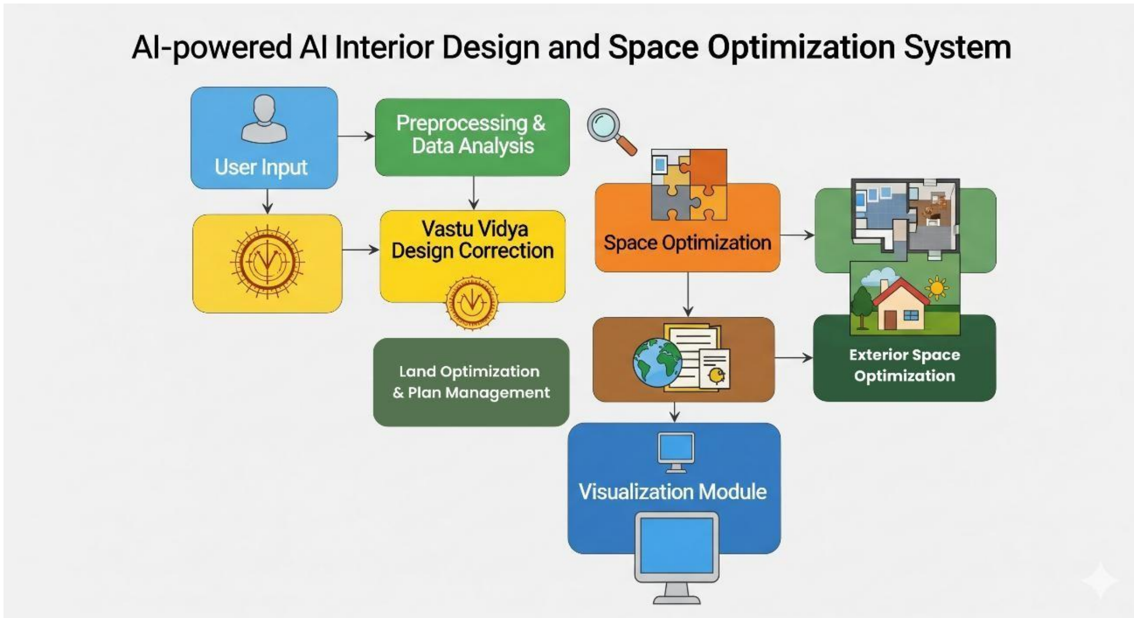

Figure 3.1 System Overview Diagram

User Input – The first step in the system is user input, where designers or homeowners enter their choices. These preferences could include cultural or environmental criteria, as well as how they would like to utilize their garden or rooftop (e.g., for gardening, recreation, or shade). Since it serves as the basis for creating customized design outputs, this input is essential.

Preprocessing & Data Analysis – The system preprocesses and analyzes the data after gathering user input. This entails sanitizing, classifying, and combining environmental variables such soil data, wind direction, and solar exposure. The system makes sure that the design process is precise and adapted to actual conditions by standardizing and evaluating this data.

Vastu Vidya Design Correction – The system now uses Wasthu Vidya, or traditional Sri Lankan design principles. These spatial and cultural adjustments guarantee that the layouts are in line with cultural identity and legacy while also being optimized for sustainability and efficiency. This brings classical architectural knowledge and contemporary AI-driven design into harmony.

Space Optimization – After that, the system uses AI-powered optimization algorithms to effectively distribute and oversee available land and outdoor areas. It makes sure that little urban plots are used efficiently by balancing uses like gardening, passive cooling, and leisure time by using geospatial analysis and optimization tools.

Exterior Space Optimization – Particularly, facades, rooftops, and gardens are the subject of this module. It guarantees that these areas are not only aesthetically pleasing but also useful by mimicking wind flow, solar shade, and microclimate control. The system offers layouts that are tailored to promote usage, increase comfort, and improve energy economy.

Visualization Module – Lastly, the algorithm generates visuals of the optimized design that are photorealistic. Using AI-based generative tools such as ControlNet and diffusion models, it enables users to see realistic previews of their outside plans. Before their plans are ever implemented, homeowners can use these visualizations to interactively assess and improve them.

3.2. Component Diagram

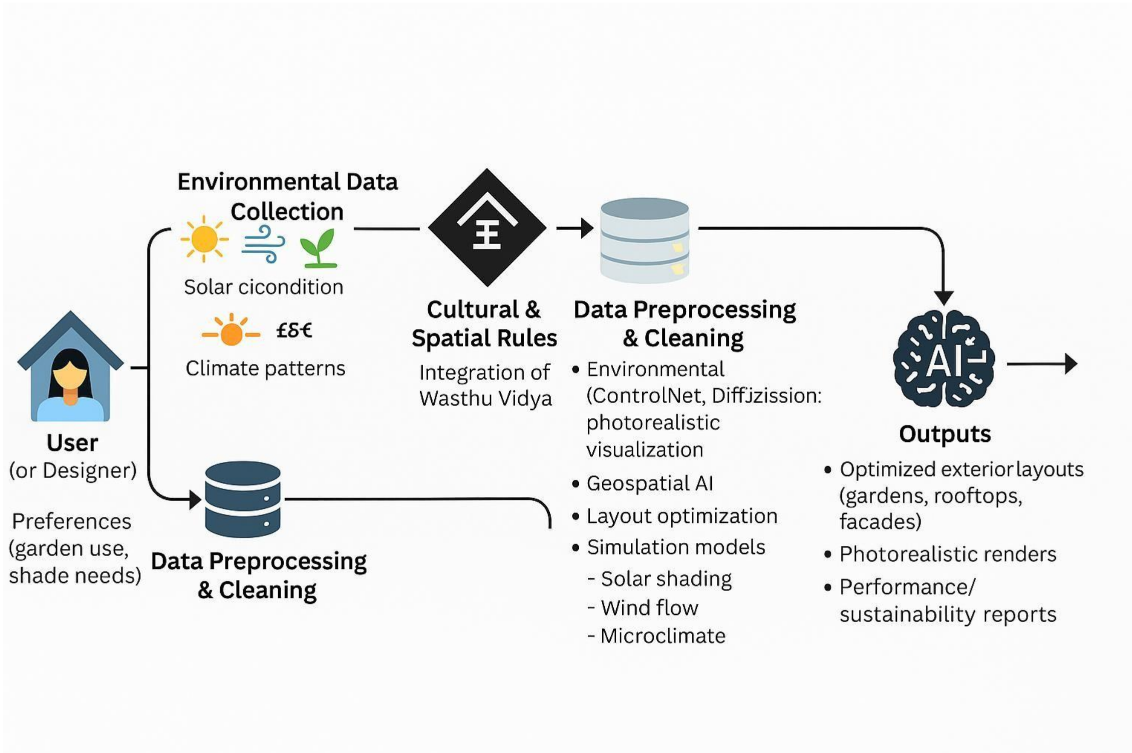

Figure 3.2 Component Diagram

User Interaction – The user (homeowner or designer) provides input at the start of the procedure. At this point, the system gathers preferences including the necessity for shaded areas, personal design choices, and how the garden, rooftop, or façade should be used. Instead of being solely system-driven, these preferences guarantee that the design is tailored and user-centered.

Environmental Data Collection – Important environmental data, such as soil quality, wind direction, solar exposure, and regional climatic patterns, are then recorded by the system. The foundational natural conditions required to maximize outdoor areas' comfort, energy efficiency, and long-term sustainability are supplied by these sources.

Cultural & Spatial Rules – Wasthu Vidya's traditional spatial ideas are incorporated throughout the design process. This makes the solutions both useful and culturally appropriate by guaranteeing that the layouts produced adhere to Sri Lankan cultural values, orientation requirements, and spatial harmony.

Data Preprocessing & Cleaning – Following collection, the user and environmental data is cleaned, verified, and standardized. By reducing inconsistencies and preparing the dataset for sophisticated AI processing, this stage guarantees the accuracy of the results.

AI-driven Optimization & Simulation – Photorealistic design previews are then produced using diffusion-based visualization technologies and sophisticated AI models like ControlNet. In order to optimize the location of trees, rooftop features, and façade design, geospatial AI and simulation models are also utilized to analyze solar shading, airflow, and microclimate performance.

Output Generation – Finally, the system produces optimized exterior layouts for gardens, rooftops, and facades. The outputs include photorealistic renders for visualization and sustainability reports that highlight improvements in natural cooling, ventilation, and space utilization. These results enable homeowners to make informed, environmentally responsible, and culturally meaningful design decisions.

3.3. Data Collection and Preprocessing

The methodology's initial step is to collect and arrange the data needed for optimization. Meteorological and urban planning authorities will provide environmental datasets, including maps of solar exposure, wind direction, rainfall intensity, and soil quality. To make sure that the designs reflect Sri Lankan values, Wasthu Vidya cultural principles and regional gardening customs will also be recorded. Surveys will also be used to collect information on user preferences, including plant types, shade needs, and recreational requirements. To facilitate seamless incorporation into AI models, all of these data sources will be preprocessed into structured formats, such as vector data for airflow and geographical heatmaps for sunshine.

3.4. AI Model Development

During this phase, simulations will be used to evaluate the AI-generated designs' environmental performance. Models of solar shading will quantify the decrease in household heat loads. Passive ventilation enhancements depending on vegetation and layout will be examined using wind flow simulations. Models of the soil and microclimate will assess drainage, humidity control, and plant compatibility. These simulations will confirm whether the designs improve outdoor comfort, lower cooling demands, increase energy efficiency, and look good.

3.5. Environmental Simulation and Modeling

During this phase, simulations will be used to evaluate the AI-generated designs' environmental performance. Models of solar shading will quantify the decrease in household heat loads. Passive ventilation enhancements depending on vegetation and layout will be examined using wind flow simulations. Models of the soil and microclimate will assess drainage, humidity control, and plant compatibility. These simulations will confirm whether the designs improve outdoor comfort, lower cooling demands, increase energy efficiency, and look good.

3.6. Real-Time Visualization Platform

To enable designers and homeowners to explore and alter layouts in real time, an interactive visualization system will be created. Users will be able to drag and drop features like trees, seating places, or rooftop panels and quickly see the lifelike result thanks to the integration of AI outputs with ControlNet-driven rendering. Alongside the visuals, environmental metrics like shade %, airflow enhancements, and energy savings will be shown. This guarantees a transparent, user-driven, and flexible decision-making process.

3.7. Evaluation and Validation

In the last step, the system is tested and validated against other AI-based landscape design platforms, including Layout2Rendering, UrbanGenAI, and PlantoGraphy. On the basis of usability, cultural integration, sustainability advantages, and environmental simulation accuracy, a comparative analysis will be carried out. To gauge acceptance, usefulness, and general happiness, user surveys including Sri Lankan planners, architects, and homeowners will be conducted. The assessment will show how the system balances sustainability, cultural identity, and contemporary efficiency while highlighting advancements over conventional design methodologies.

4. DESCRIPTION OF PERSONNEL AND FACILITIES

4.1. Personnel

The student researcher will conduct this study as a component of their undergraduate project. The pupil is accountable for:

- reviewing literature and determining the area of study that needs further attention in AI-driven external space optimization.
- gathering and preprocessing pertinent information about user preferences, culture, and the environment.
- creating and putting into practice AI optimization models that incorporate generative design and geographical analysis.
- constructing the microclimate, wind flow, and sun shading simulation models to assess environmental performance.
- creating and putting into use the real-time visualization interactive user interface.
- recording the results of the study and testing, validating, and contrasting the system with current solutions.

The designated academic supervisor will supervise and guide the project, reviewing the technical design, offering methodological feedback, and guaranteeing academic rigor throughout its execution.

4.2. Facilities

The initiative will mostly make use of the university's computer and lab resources. Resources that are needed include:

1. Access to a computer lab: For simulation work, AI model training, and software development.
2. Software Tools: Grasshopper/Blender (for parametric modeling), TensorFlow/PyTorch (for AI modeling), Python (for AI and simulation), and rendering tools (ControlNet, Stable Diffusion).
3. Data Sources: Open-source datasets pertaining to Sri Lankan urban climate, soil quality, wind patterns, and solar radiation.
4. Access to the Internet and libraries: For research publications, dataset retrieval, and literature reviews.
5. Testing Facilities: Computer resources from the university to execute environmental simulation models and AI training.

To manage computationally demanding AI model training and photorealistic rendering tasks, extra resources like cloud-based GPU services (Google Colab, AWS, or universityprovided HPC clusters) will be used if necessary.

References

[1] R. Xing, M. Li, and P. Zhang, "Artificial Intelligence in Landscape Architecture: A Survey," *Landscape and Urban Planning*, vol. 245, p. 104634, 2024.

[2] H. Wang, S. Lin, and T. Zhao, "PlantoGraphy: Iterative generative design for landscape visualization," *ACM Transactions on Graphics*, vol. 43, no. 2, pp. 1–12, 2024.

[3] J. Zhou and Q. Li, "UrbanGenAI: Generative AI for urban landscape reconstruction," *Computers, Environment and Urban Systems*, vol. 107, p. 101932, 2024.

[4] Y. Chen, L. Zhao, and J. Wu, "Layout2Rendering: AI-aided greenspace design through deep learning and Grasshopper integration," *Journal of Computational Design*, vol. 12, no. 3, pp. 105–121, 2024.

[5] F. Ashari, J. Müller, and K. Becker, "Customizable AI tools for landscape rendering: A comparison with traditional workflows," *Proceedings of Digital Landscape Architecture*, vol. 26, no. 1, pp. 15–25, 2025.

[6] K. Perera, *Wasthu Vidya: Traditional design principles in Sri Lankan housing*. Colombo: National Heritage Publications, 2019.

[7] D. Fernando and T. Samarasinghe, "Urban housing and the decline of traditional outdoor spaces in Sri Lanka," *Journal of Architecture and Environment*, vol. 18, no. 2, pp. 44–57, 2021.